BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: Data Representation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does the MDR hold?

2. grid = [[11,12],[21,22]]. With indexes starting at 0, what is grid[1,0]?

A) 21 B) 11
C) 12 D) 22

3. "Changing a variable name automatically fixes every logic error." Correct this. Explain why.

4. Two functions each have a local total. Several procedures share a global score. Correct both claims. Explain why. (a) All subprograms can read every local variable. (b) Only one subprogram can read a global variable.

Today's objectives

Explain the core ideas of paper 1: data representation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Binary place values

Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers.

Hexadecimal

Group binary digits into nibbles of four bits.

Image size

Width x height x colour depth yields uncompressed bits.

Sound size

Samples/second x seconds x bits/sample yields mono uncompressed bits.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Binary place values	Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers.
Hexadecimal	Group binary digits into nibbles of four bits.
Image size	Width x height x colour depth yields uncompressed bits.
Sound size	Samples/second x seconds x bits/sample yields mono uncompressed bits.

I Do - One worked example

Convert eight-bit 10110110 to denary and hexadecimal.

1. Add values: 128+32+16+4+2=182.
2. Split 1011 and 0110.
3. Convert to B6 hexadecimal.
4. Check 11 x 16+6=182.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Calculate the bits in a 60 x 40 bitmap with four-bit colour depth.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the raw data size conversion from bits to bytes.

```
width = 20
height = 10
depth = 8
bytes = width * height * depth
print(bytes)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Convert hex 2A to denary. [3 marks]

Q3. Convert eight-bit binary 11001100 to hex. [2 marks]

Q4. Calculate mono sound bytes at 1000 samples/s, 4 s, 8 bits/sample. [4 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. One hexadecimal digit can represent four binary bits. ☐ T ☐ F
2. Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers. ☐ T ☐ F
3. Group binary digits into nibbles of four bits. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Data or an instruction being transferred to or from memory. ☐

Q2 A - 21 ☐

Q3 Clear names show what values mean and help people read and change code. ☐

Q4 (a) Each local total is used only in its own scope. (b) A global score can be used across the program. This risks unwanted changes. Use arguments and return values to make changes clear. ☐

Code First - Model

```
width = 20
height = 10
depth = 8
bytes = width * height * depth DIV 8
print(bytes)
```

Output: 200

We Do - Check the model

1. Divide eight to get 1200 bytes if requested. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 2 x 16+10=42. ☐

Q3 CC. ☐

Q4 1000 x 4 x 8 / 8=4000 bytes. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: